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Pre-set boundaries

- 2000–2020
- All individuals who completed the individual questionnaire at least once (n = 90,357)
- Access to: pl, hl, biol, biobirth, bioparen, pgen, hgen, and ppathl. Note: There seems to be no access to phrf, which is the file with weights.

Hypotheses

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

H2. Having daughters leads to more left-leaning political preferences.

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

General exclusion criteria for all hypotheses

- Exclude observations with no first child in period of observation
- Exclude observations where sex of child is unknown

Hypothesis 1: gender equality attitudes

Construction of dependent variable - genderAtt

Gender attitudes will be measured using the variables plj0621:plj0625. These questions were asked in 2016, 2017, and 2018 on a scale of 1-7 (1=completely disagree, 7=completely agree).

The items from the battery of questions on the position of women are as follows:

plj0621	Having a job is the best way for a woman to be independent.
plj0622	Even a married woman should have a paid job so that she can be financially independent.
plj0623	If a woman earns more money than her partner, this inevitably leads to problems.
plj0624	For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters.
plj0625	At home, the husband should have the final say.

In the original coding, 1 = disagree and 7 = agreement.

plj0623, plj0624 and plj0625 will be reverse coded. A higher score then stands for more gender equality and a lower score for lower gender equality.

I will then check Cronbach's alpha to see which items are internally consistent.

A Cronbach's alpha of >=0.7 is considered sufficient internal consistency.

If Cronbach's alpha is <=0.7, I will identify the alpha with a stepwise removal of each item, to identify which item might cause low internal consistency. I will use the constellation of items that has a Cronbach's alpha >=0.7 and the highest number of items. (e.g., if 2 items have a Cronbach's alpha of 0.75 and three items have 0.72, I will use the three items, not the two).

Using the selected items, I will construct a mean index of the items to create the variable genderAtt.

Construction of main independent variable (predictor variable) – hadDaughter

Identification of individuals who had a daughter (vs. who only had sons) in the period between first and last interview using BIOBIRTH dataset.

1) kidsex01F: recode kidsex01 so that 0 = 1, 1 = 2 (female = 1, male = 0)

2) if kidsex01F == 0, hadDaughter = 0, if sum of kidsex01F == 1, hadDaughter = 1

The variable hadDaughter will be 1 if the child is the first daughter, and 0 if the first child is a son.

Construction of other variables

birthOrder = can be identified using kidpnr01–kidpnr19 (persnr child), and kidgeb01– kidgeb19 (year of birth)

syear already exists in the data.

Inclusion/exclusion criteria

- Exclude observations with missing value on plj0621- plj0625 in any year in the period 2016-2018
- Exclude observations without observation of first childbirth in either 2017 or 2018
    - Gender attitudes measured in 2016, 2017, 2018, so to capture attitudes after birth.
- Exclude observations with unknown sex of child
- Only include individuals with measurement of gender attitudes before and after childbirth
- caution, expected sample size is small
- as explained below, some observations are dropped through the model specification

Model construction

I will use a fixed effects regression with clustered standard errors on the individual level.

Dependent variable: genderAtt

Independent variable: hadDaughter

Fixed effects: respondent, survey year, birthOrder

Model specification in stata:

xtset pid sysear

xtreg genderAtt hadDaughter i.birthOrder i.syear, fe vce(cluster pid)

No controls: Classic sociodemographic controls (marriage, income, education etc). impact whether you have a son or daughter. It would be important to control for confounders, i.e. factors that influence the treatment and the outcome. However, it is unlikely that there are observed or unobserved factors that influence both the infant's sex and gender attitudes.

The year fixed effects account for time trends and also captures annual shocks.

If the sample size is small the year fixed effects might need to be dropped. This would be unproblematic as the cohort effect itself is likely small. However, if possible it is included as there may be annual shocks or trends.

This model uses parent fixed effects, so relies on variation within each parent. Any parent who does *not* have a female birth after the first interview and before the last interview (i.e. the treatment is 0-0-0 for the three years, or 1-1-1 if they had a daughter before the first interview), is dropped from the regression. This means that I am essentially identifying the effect of having a female birth only among those who have a female birth (0-1-1 or 0-0-1), i.e., a before and after. Essentially then I am asking, for those who do have a female birth in the observational period, do their gender attitudes change.

Interpretation of results

I set α=0.05 and will reject the null when the p-value is less than 0.05.

Hypothesis 2: Left-leaning political preferences

Construction of dependent variable – leftVote

Left-leaning political preferences will be based on party that a person would vote for. I will use the harmonised variable plh0012_h available in the pl dataset. It is coded as 1= SPD, 2 = CDU, 3 = CSU, 4 = FDP, 5 = Buendis90/Gruene, 6 = Die Linke, 7 = NPD/Republikaner/ Die Rechte, 8 = Andere, 9 = SPD/Buendnis90/Gruene, 10 = SPD/CDU, 11 = SPD/FPD, 12 = SPD/DVU(Rep/NPD), 13 = CDU/CSU, 14 = CDU/FDP, 15 = CDU/Buednis90/Gruene, 16 = Buednis90/Gruene/Die Linke, 17 = SPD/Die Linke, 18 = Gruene/DVU(Rep/NPD), 19 = Die Linke/ DVU(Rep/NPD), 20 = CDU/Die Linke, 21 = CDU/ DVU(Rep/NPD), 22 = CSU/FDP, 23 = Gruene/FDP, 24 = FDP/Die Linke, 25 = FDP/NPD, 26 =Piratenpartei, 27 =AfD, 28 = did not vote, 29 = was not eligible to vote, 30 = CDU+/or CSU/AfD, 31 = SPD/AfD. -1 to -8 are missings.

I will recode plh0012_h to create left_vote (1= 1, 5, 6, 9, 16, 17; 0 = 2, 3, 4, 7,8, 10, 11, 12, 13, 14, 15, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 30, 31). 28 and 29 will be coded as missing as they did not vote.

This variable has the best coverage compared to alternatives and is in line with previous research e.g. https://doi.org/10.1177/00104140241271123.

Construction of main independent variable (predictor variable) – hadDaughter

Same as for hypothesis 1.

Construction of other variables

Same as for hypothesis 1.

Inclusion/exclusion criteria

Analogously to hypothesis 1, only include individuals with measurement of political attitudes before *and* after childbirth

Model construction

Dependent variable: leftVote

Independent variable: hadDaughter

Fixed effects: pid, survey year, birthOrder

Model specification in stata:

xtset pid sysear

xtreg leftVote hadDaughter i.birthOrder i.syear, fe vce(cluster pid)

Interpretation of results

I set α=0.05 and will reject the null when the p-value is less than 0.05.

Hypothesis 3: gender equality in the household

Construction of dependent variable – careRatio

Measured as the ratio of care work within a household between the mother and father.

Variable creation:

1) Identify cohabiting couple that had a child during SOEP data collection, based on most recent year before childbirth.
2) Combine measure of housework, care work, and childcare for mother and father.
    a. pli0043_h, pli0046, pli0044_h. Coverage for all years 2000–2020 apart from pli0046 which was not asked 2000. -1 to -8 are missings. Variable is the number of hours per average weekday.
3) Calculate ratio of mother's to father's care work
    a. mother(pli0043_h + pli0046 + pli0044_h)/father(pli0043_h + pli0046 + pli0044_h)

Example 1:

- Mother: 8 hours housework, 4 hours care work, 10 hours childcare = 22 hours total
- Father: 6 hours housework, 0 hours care work, 0 hours child care = 6 hours total
- Mother:Father ratio = 22:6 = 3.67 i.e. mother does 3.67 times more care work than father

Example 2:

- Mother: 8 hours housework, 4 hours care work, 10 hours childcare = 22 hours total

- Father: 8 hours housework, 4 hours care work, 10 hours child care = 22 hours total
- Mother:Father ratio = 22:22 = 1 i.e. mother does the same amount of care work as father

Example 3:

- Mother: 8 hours housework, 4 hours carework, 10 hours childcare = 22 hours total
- Father: 12 hours housework, 3 hours carework, 15 hours child care = 30 hours total
- Mother:Father ratio = 22:30 = 0.73 i.e. mother 0.73 of the care work that the father does

Construction of main independent variable (predictor variable) – hadDaughter

Same as for hypothesis 1.

Construction of other variables

Same as for hypothesis 1.

Inclusion/exclusion criteria

- Inclusion: household of cohabiting parents (identified using hid, pgpartnr, pgpartz, mnr, fnr as needed)
    - Only couples that cohabited at least one year before childbirth (to have a first measurement before pregnancy)
- Inclusion: full observations for care work variables pli0043_h, pli0046 or pli0044_h in a given year (e.g. if 2015 is complete, but 2016 is not, 2016 would be dropped from the analysis and the predictor for the couple would get NA for that year.)
- Only include individuals with measurement of careRatio before and after childbirth

Model construction

Dependent variable: careRatio

Independent variable: hadDaughter

Fixed effects: pid, survey year, birthOrder

Model specification in stata:

xtset pid sysear

xtreg careRatio hadDaughter i.birthOrder i.syear, fe vce(cluster pid)

Interpretation of results

I set $\alpha=0.05$ and will reject the null when the p-value is less than 0.05.

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Hypothesis 4: gender equality attitudes

Construction of dependent variable – genderAtt

Same as for hypothesis 1.

Construction of main independent variable (predictor variable) – hadDaughter

Same as for hypothesis 1.

Construction of control variables

Same as for hypothesis 1.

Inclusion/exclusion criteria

- Same as hypothesis 1.
- Same caution: small sample size expected

Model construction

Dependent variable: genderAtt

Independent variable: hadDaughter*female – interaction term

Fixed effects: pid, survey year, birthOrder

Model specification in stata:

xtset pid sysear

xtreg genderAtt hadDaughter##female i.birthOrder i.syear [pw=wlong], fe vce(cluster pid)

Interpretation of results

I set $\alpha=0.05$ and will reject the null when the p-value is less than 0.05.